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Superlattices and Microstructures

journal homepage: www.elsevier.com/locate/superlattices

Impact of line edge roughness on the performance of 14-nm FinFET: Device-circuit Co-design

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ARTICLE INFO

Article history:

Received 5 September 2017

Received in revised form 31 October 2017

Accepted 31 October 2017

Available online xxx

Keywords:

FinFET

Line edge roughness

Resist-defined

Spacer-defined

SRAM

Variability

ABSTRACT

With the evolution of sub-20 nm FinFET technology, line edge roughness (LER) has been identified as a critical problem and may result in critical device parameter variation and performance limitation in the future VLSI circuit application. In the present work, an analytical model of fin-LER has been presented, which shows the impact of correlated and uncorrelated LER on FinFET structure. Further, the influence of correlated and uncorrelated fin-LER on all electrical performance parameters is thoroughly investigated using the three-dimensional (3-D) Technology Computer Aided Design (TCAD) simulations for 14-nm technology node. Moreover, the impact of all possible fin shapes on threshold voltage (V_{TH}), drain induced barrier lowering (DIBL), on-current (I_{ON}), and off-current (I_{OFF}) has been compared with the well calibrated rectangular FinFET structure. In addition, the influence of all possible fin geometries on the read stability of six-transistor (6-T) Static-Random-Access-Memory (SRAM) has been investigated. The study reveals that fin-LER plays a vital role as it directly governs the electrostatics of the FinFET structure. This has been found that there is a high degree of fluctuations in all performance parameters for uncorrelated fin-LER type FinFETs as compared to correlated fin-LER with respect to rectangular FinFET structure. This paper gives physical insight of FinFET design, especially in sub-20 nm technology nodes by concluding that the impact of LER on electrical parameters are minimum for correlated LER.

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1. Introduction

Recently, with the advancement of integrated circuits (ICs) manufacturing process, the feature size of conventional MOS transistor shrinks down to a nanometer regime. This aggressive scaling of conventional MOS transistor will introduce various short channel effects (SCEs) [1,2]. To alleviate these SCEs in nanometer regime, various alternative planar and non-planar device structures have been proposed [3–7]. Among them, the FinFET structure has found to be the most capable contender due to its better electrostatics and excellent compatibility with the standard CMOS technology [5–7]. However, the introduction of non-planar 3-D channel region will result in non-uniform fin width and random inclined side wall surfaces due to limitation of the fabrication process and etching steps [8–11]. Consequently, there will be a random fin shape fluctuation along the horizontal and vertical direction. This results in irregular fin shape instead to rectangular fin and degrades the FinFET performance significantly. These fluctuations commonly known as line edge roughness (LER) variability, and this

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pose a critical problem in the performance of the FinFET device because LER doesn't scale down with technology node [12]. LER mainly occurs due to slower dissolution rate of large polymer aggregates in the removal process of undesirable photoresist [13,14]. The rough edge of photoresist is transferred to physical patterned lines on the semiconductor, such as gate and fin patterning. Recently, many authors discussed the influence of LER and non-ideal geometry of FinFET structure [15–23]. But, most of the work demonstrates the impact of non-rectangular fin shape (wider from the bottom and thinner from the top) like convex, concave, trapezoidal etc. in the vertical fin direction [16–18]. However, there are very few studies concerning the variations in lateral direction [19–23], whereas these kinds of fluctuations also exist in photolithographic process step due to LER [8,9]. These geometric fluctuations along fin width are responsible for variation in electrostatic control of fin. Consequently, device performance becomes random and unpredictable. This ultimately leads to a statistical distribution in parameters such as threshold voltage (V_{TH}), drain induced barrier lowering ($DIBL$), on-current (I_{ON}), and off-current (I_{OFF}) characteristics. This can be more troublesome in circuit applications, especially in SRAM where precise device matching is critically required [22–24].

In the present work, an analytical model for fluctuations introduced by fin-LER has been presented, which illustrates the influence of correlated and uncorrelated LER on fin shape. Further, all possible non-rectangular fin shapes resulted from spacer-(i.e. correlated LER) and-resist (i.e. uncorrelated LER)-defined patterning techniques have been explored for 14-nm FinFET device. In addition, the impact of all possible fin shapes on read static noise margins (RSNM) performance of FinFET based 6-T SRAM has been investigated. To get better insight, all the results are compared to well calibrated rectangular FinFET structure. This analysis is very helpful to predict the influence of non-rectangular fin shape on the device as well as circuit level variability efficiently. The rest of article is structured as follows. Section 2 demonstrates the nominal FinFET device design and all possible fin shapes. Simulation results have been explained in Section 3. Finally, the conclusion of the paper has been drawn in Section 4.

2. Simulation setup and model formulation

2.1. Nominal FinFET design and calibration

Fig. 1 shows the schematic cross-sectional view of nominal rectangular FinFET structure. All the geometrical and doping parameters of a rectangular FinFET (i.e. without LER) has been shown in Table 1. For spacer and gate oxide material, high- k (i.e. HfO_2) and low- k (i.e. SiO_2) material has been considered, respectively. To control short channel effects (SCEs) and for optimum subthreshold performance the Gaussian graded doping profile decaying over the gate-source/drain (gate-s/d) underlap length of 6 nm has been taken [25,26]. The gate oxide thickness layer (t_{ox}) and silicon fin thickness (T_{si}) plays a crucial role in controlling the carrier mobility as well as threshold voltage (V_{TH}). In the present work, T_{si} of 8 nm and t_{ox} of 1 nm has been considered to efficiently suppress the effect of SCEs. TiN mid-gap metal gate electrodes have been considered due to fact that it has a high melting point, good thermal stability, low resistivity and most important TiN shows excellent compatibility with standard CMOS process [27]. All the simulations in this paper are carried out using Synopsys Sentaurus TCAD 3-D simulator [28]. To capture the quantum confinement effects of inversion carriers in thin body the quantum potential model has been turned on. The high- k Lombardi mobility model has been activated to account for high- k mobility degradation at the semiconductor-insulator interface.

Before analyzing the impact of LER on a 14-nm FinFET structure, it needs a careful calibration in respect of real fabricated nanoscale FinFET structure. The test-bed FinFET structure having a physical gate length of 14-nm is successfully fabricated by Intel in 2014 [29]. The I_{OFF} and I_{ON} values have been reported for the fabricated device. This calibration aims to closely replicate the physical dimensions, and doping profiles in TCAD Synopsys simulation and to match consistently the measured electrical characteristics [29]. The main purpose of this calibration is to closely replicate the physical dimensions and doping profiles in TCAD Synopsys simulation in order to match the measured electrical characteristics. The calibration can be done by matching the device electrostatics with an aim to match the same sub-threshold (SS) and $DIBL$. Thereafter, mobility has been tuned to match the current level. Finally, I_D - V_{GS} and I_D - V_{DS} characteristics of our FinFET device has been calibrated with Intel 14-nm FinFET device. Similarly, p -type 14-nm FinFET structure is calibrated. Figs. 2 and 3 illustrate the calibrated transfer-and output-characteristics of the n - and p -channel FinFET, respectively. The key figures of merit extracted from simulations are

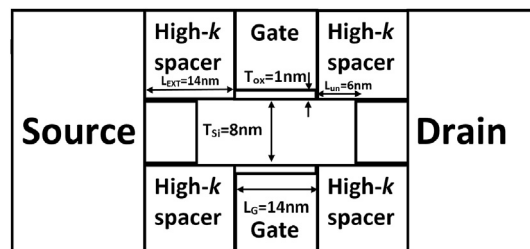
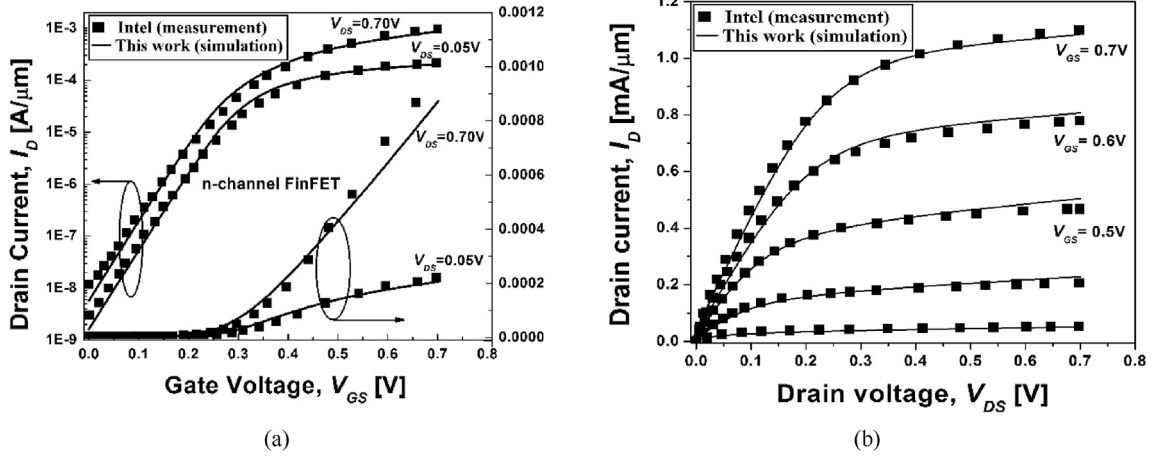
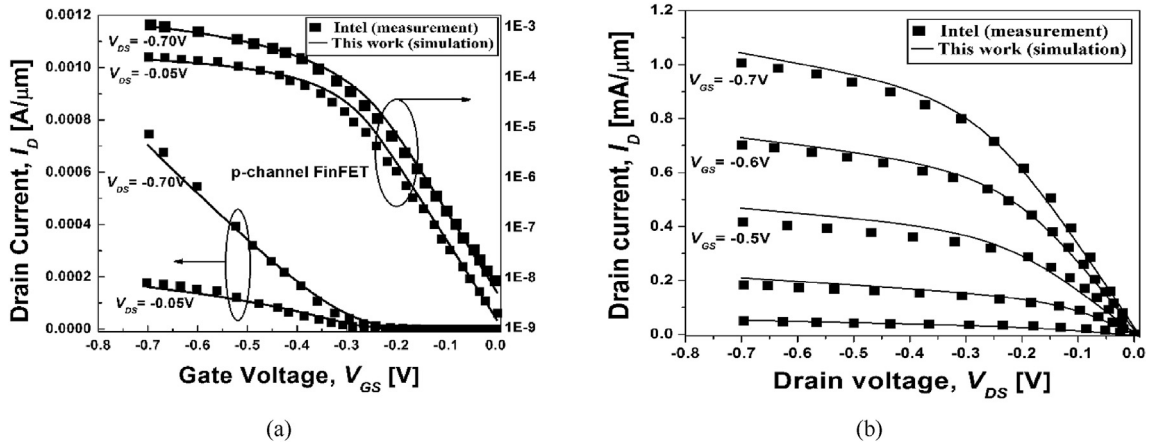


Fig. 1. The schematic of cross-sectional view of the nominal 14-nm FinFET device.

Table 1

Simulated nominal rectangular FinFET device parameters.

Parameter	Value
Fin thickness (T_{si}) [nm]	8
Fin height (H_{fin}) [nm]	20
Physical gate length (L_G) [nm]	14
Gate oxide thickness, (t_{ox}) [nm]	1
S/D extension spacer length, (L_{EXT}) [nm]	14
Channel doping (N_A) [cm^{-3}]	1×10^{16}
S/D doping (N_D) [cm^{-3}]	1×10^{20}

**Fig. 2.** Calibration of (a) transfer (I_D - V_{GS}) and (b) output (I_D - V_{DS}) characteristics with fabricated Intel 14-nm gate length *n*-channel FinFET structure.**Fig. 3.** Calibration of (a) transfer (I_D - V_{GS}) and (b) output (I_D - V_{DS}) characteristics with fabricated Intel 14-nm gate length *p*-channel FinFET structure.

compared with fabricated data in Table 2. It is clear from Figs. 2 and 3, that there is a good agreement between transfer-and output-characteristics of experimental and simulated results. This grants confidence that Synopsys simulator is successfully able to imitate the behavior of the Intel 14-nm FinFET structure.

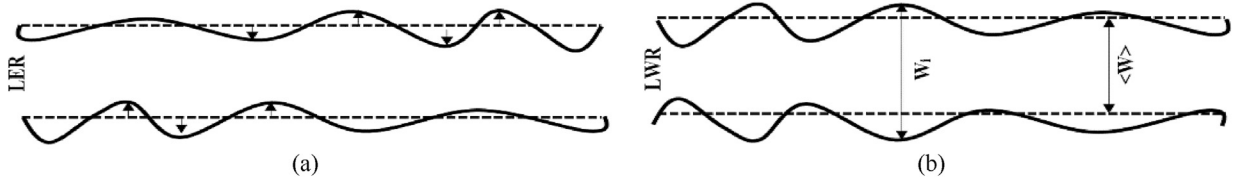
2.2. Model formulation for fin LER

In FinFET structure, LER is responsible for overall body thickness fluctuations, which significantly affects the device performance parameters in sub-20 nm technology node. LER comes into picture mainly because of fabrication limitation of lithography and etching process [8,9,30,31]. This will cause fluctuations in lateral and vertical direction of the fin in FinFET structure. Now, it is important to distinguish between line width roughness (LWR) and line edge roughness (LER). So, Fig. 4

Table 2

Simulated and fabricated performance parameters of FinFET structure.

Parameter	Intel (Fabricated)	This work (Simulated)
ON-current (I_{ON}) [$\mu\text{A}/\mu\text{m}$]	1100	1024
OFF-current (I_{OFF}) [$\eta\text{A}/\mu\text{m}$]	10	3.62
Subthreshold slope (SS) [mV/dec]	65	65.5
DIBL [mV/V]	60	20.98

**Fig. 4.** Definition of LER and LWR in a line pattern. (a) LER is fluctuation of real edge position (solid line) about its average value (dotted line) (b) LWR is deviation of line width along the line pattern.

depicts the behavior of LER and LWR profile patterns. It is clear from Fig. 4 (a) that LER denotes the variations in both sidewall line pattern of the fin about its average value, whereas as shown in Fig. 4 (b) the deviations in line width to the average value to fin thickness is commonly known as LWR [32]. LWR is given by the variation of the line width as below,

$$\Delta_{LWR}^2 = \frac{1}{N-1} \sum_{i=1}^N (W_i - \langle W \rangle)^2 \quad (1)$$

where, Δ_{LWR} describes the magnitude of fin width roughness, N is the number of sample along the length of fin sidewall, W_i is width of i th sample, and $\langle W \rangle$ is the mean value of fin thickness. The correlation between the individual left-and right edges of the fin and LWR can be given as shown below,

$$\Delta_{LWR}^2 = \Delta_L^2 + \Delta_R^2 - 2\rho_X \Delta_L \Delta_R \quad (2)$$

where, Δ_R and Δ_L are the fluctuation in right and left edges of the fin, respectively, and ρ_X denotes the cross-correlation coefficient between the left and right edges of fin. The value of ρ_X completely depends on the fabrication process and it varies between -1 and $+1$, where -1 signifies anti correlation, 0 means no correlation, and $+1$ signify perfect correlation between the left and right edges of the fin. If we assume the same amount of fluctuation between the left and right edges of the fin i.e. $\Delta_L = \Delta_R = \Delta_{LER}$ then magnitudes of LER and LWR can be given by simplifying Equation (2) as below,

$$\Delta_{LWR}^2 = 2\Delta_{LER}^2(1 - \rho_X) \quad (3)$$

thus, from Equation (3) if $\rho_X = 1$, then $\Delta_{LWR} = 0$. It results in perfect correlation between the left-and right edges of the fin even if Δ_{LER} is non-zero. This means that the overall thickness of the fin doesn't affect due to LER. The perfect correlation between the two edges of the fin can be achieved by using the spacer defined patterning rather than conventional resist defined patterning technique.

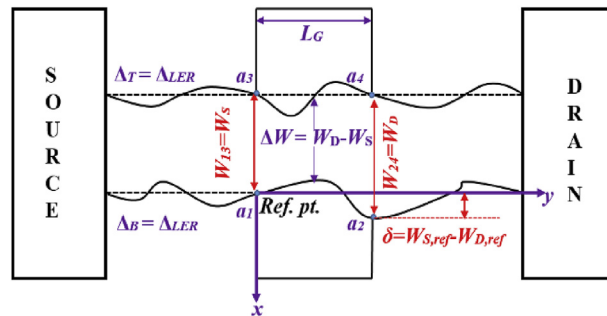
**Fig. 5.** Fin shape fluctuations due to LER and description of model parameters. Source and drain sides are assumed to be left and right position, respectively. Point a_1 - a_4 are locations where the gate electrode intersects the fin.

Fig. 5 illustrates the basic model parameters used to formulate the statistical variations due to LER in FinFET structure. Here, a_1, a_2, a_3, a_4 are random variables at which this point the fin and gate electrode intersects each other, $\mu_1, \mu_2, \mu_3, \mu_4$ are mean or average values, $\Delta_{12}, \Delta_{22}, \Delta_{32}, \Delta_{42}$ are the variances between two random points, and $\rho_{12}, \rho_{13}, \rho_{14}, \rho_{23}, \rho_{24}, \rho_{34}$ are correlation coefficient of respective random variables. Δ_T and Δ_B are the top and bottom fin edge fluctuation, respectively. The relationship between mean and variance of any correlated random variables (A) is given by a linear combination, described by Equation (4) [15],

$$A = \psi_1 a_1 + \psi_2 a_2 + \psi_3 a_3 + \psi_4 a_4 = \sum_{i=1}^n \psi_i a_i \quad (4)$$

where, ψ_1, ψ_2, ψ_3 , and ψ_4 are any positive or negative constants of a_1, a_2, a_3 , and a_4 , respectively, and expectation (E) of correlated random variables is given by Equation (5),

$$E(A) = \psi_1 \mu_1 + \psi_2 \mu_2 + \psi_3 \mu_3 + \psi_4 \mu_4 = \sum_{i=1}^n \psi_i \mu_i \quad (5)$$

And variance, $V(A)$ will have “ n -squared” terms and cross-correlation terms,

$$\begin{aligned} \Delta_\delta^2 = V(A) &= \psi_1^2 \Delta_1^2 + \psi_2^2 \Delta_2^2 + \psi_3^2 \Delta_3^2 + \psi_4^2 \Delta_4^2 + 2\psi_1 \psi_2 \Delta_1 \Delta_2 \rho_{12} + 2\psi_1 \psi_3 \Delta_1 \Delta_3 \rho_{13} + 2\psi_1 \psi_4 \Delta_1 \Delta_4 \rho_{14} \\ &+ 2\psi_2 \psi_3 \Delta_2 \Delta_3 \rho_{23} + 2\psi_2 \psi_4 \Delta_2 \Delta_4 \rho_{24} + 2\psi_3 \psi_4 \Delta_3 \Delta_4 \rho_{34} = \sum_{i=1}^n \psi_i^2 \mu_i^2 + 2 \sum_{i=1}^n \sum_{j=i+1}^n \psi_i \psi_j \Delta_i \Delta_j \rho_{ij} \end{aligned} \quad (6)$$

where, i and j are line width roughness profile points between the edges of fin, and ρ_{ij} denotes the cross-correlation between i and j points. Now, an offset parameter δ as the fluctuation between the bottom edge of drain and source side has been considered, namely point a_1 and a_2 shown in Fig. 5.

$$\delta = W_{S,ref} - W_{D,ref} = (a_2 - a_1) \quad (7)$$

Now from Equation (6),

$$\Delta_\delta^2 = V(a_2 - a_1) = \psi_1^2 \Delta_1^2 + \psi_2^2 \Delta_2^2 + 2\psi_1 \psi_2 \Delta_1 \Delta_2 \rho_{12} \quad (8)$$

substituting $\psi_1 = -1, \psi_2 = +1, \Delta_1 = \Delta_2 = \Delta_{LER}$, and $\rho_{12} = \rho_A(L_G)$. Here, $\rho_A(L_G)$ is autocorrelation term calculated at lag L_G . We can define offset line variations as,

$$\Delta_\delta^2 = 2\Delta_{LER}^2 [1 - \rho_A(L_G)] \quad (9)$$

So, it is clear that the gate length governs the sampling size in the auto-correlated LWR function along each individual edge of the fin. Now, the difference in fin width (δW) between the source and drain sides is given by,

$$\delta W = W_D - W_S = [(a_2 - a_4) - (a_1 - a_3)] \quad (10)$$

In Equation (10) the random but correlated variable points a_2, a_3 , and a_4 are relative to point a_1 . Again, using Equation (6),

$$\Delta_{\delta W}^2 = (\psi_2^2 \Delta_2^2 + \psi_4^2 \Delta_4^2 + 2\psi_2 \psi_4 \Delta_2 \Delta_4 \rho_{24}) - (\psi_1^2 \Delta_1^2 + \psi_3^2 \Delta_3^2 + 2\psi_1 \psi_3 \Delta_1 \Delta_3 \rho_{13}) \quad (11)$$

substituting $\psi_1 = -1, \psi_2 = +1, \psi_3 = +1, \psi_4 = -1, \Delta_1 = \Delta_2 = \Delta_3 = \Delta_4 = \Delta_{LER}$, and

$$\begin{aligned} \rho_{13} &= \rho_{24} = \rho_X(0) \\ \rho_{14} &= \rho_{32} = \rho_X(L_G) \\ \rho_{2} &= \rho_{34} = \rho_A(L_G) \end{aligned} \quad (12)$$

where, $\rho_X(0)$ and $\rho_X(L_G)$ are the cross-correlation terms of random variables calculated at 0 and L_G lag between the top and bottom edges of the fin. Now, for resist-defined patterning technique, top and bottom edges of the fin fluctuate in uncorrelated manner, therefore the cross-correlation terms $\rho_X(0) = \rho_X(L_G) = 0$, and the fluctuation in δW is given by using Equation (11),

$$\Delta_{\delta W}^2 = 4\Delta_{LER}^2 [1 - \rho_A(L_G)] \quad (13)$$

It is clear from Equations (9) and (13) that, for the same value of Δ_{LER} , the fluctuations in δW is twice. Similarly, in case of spacer-defined patterning technique fin edges fluctuates in a perfect correlated manner leading to $\rho_X(0) = 1$, and $\rho_X(L_G) = \rho_A(L_G)$. Thus, the variations in δW is zero for spacer-defined patterning technique.

$$\Delta_{\delta W}^2 = 0 \quad (14)$$

The overall variability in performance parameters (P) (i.e. V_{TH} , I_{ON} , I_{OFF} etc.) of FinFET structure is governed by the fin geometries and fin LWR generally affects the FinFET performance parameters by varying the average fin width in the channel region. Thus, the fluctuations in performance parameters P due to fin LWR can be modeled as [15],

$$\Delta_{P,fin}^2 = \left(\partial P / \partial W \right)^2 \Delta_{\delta W,fin}^2 \quad (15)$$

From Equation (15), it can be found that ideal variations in FinFET performance parameters only signify the resist-defined patterning technique due to uncorrelated LER. But, in practice spacer-defined patterning technique is not completely immune to these parameter variations.

2.3. Possible fin shapes due to resist-and spacer patterning techniques

The fin line edge roughness (fin-LER), which is inherent due to various patterning techniques is found to be one of the most dominant sources of statistical variability [19,20]. With down scaling the impact of LER increases rapidly which leads to fluctuations in various geometrical and electrical parameters thus affecting transistor operation. The resist-and spacer-defined patterning techniques have been extensively used to fabricate vertical fin of FinFET and gate electrode structures for ultrathin body MOSFETs [8–11,30]. However, as shown in Fig. 6, both resist-and spacer-defined patterning techniques are affected by LER which is responsible for fin shape fluctuations [15,19–24,30]. In resist-defined patterning technique, the resist material is responsible for random uncorrelated LER (i.e. $\Delta W \neq 0$ and $\rho_x = 0$), which results in rough resist lines on the both edges of the fin, as shown in Fig. 6 (a). Although, by combining the benefits of spacer along with resist, spacer-defined patterning results in-phase correlation (i.e. $\Delta W = 0$ and $\rho_x = 1$) between line edges and lower the effect of LER on both the edges of fin as shown in Fig. 6 (b). Hence, resist-defined patterning has fin width fluctuations along the fin length while spacer-defined patterning does not have such fluctuations.

Consequently, as illustrated in Fig. 7, fin fluctuates in the correlated manner in spacer-defined technique which results in two possible fin shapes, i.e., bent and curve, as shown in Fig. 7 (b) and 7 (c). On the other hand, fin width fluctuations take place due to uncorrelated LER in resist-defined patterning technique resulting in four possible fin shapes, i.e. big source (bigS), big drain (bigD), fat, and thin, as shown in Fig. 7 (d)–(g) [21]. It is extremely necessary to get a smaller value of LER. In the present work, LER variation of ± 3 nm for resist-and spacer-defined fin shapes has been considered as per experimental data [8–11] and ITRS projections [33]. It is important to mention here, for the sake of simplicity only low frequency LER induced fin roughness has been considered throughout the paper while ignoring the high frequency roughness.

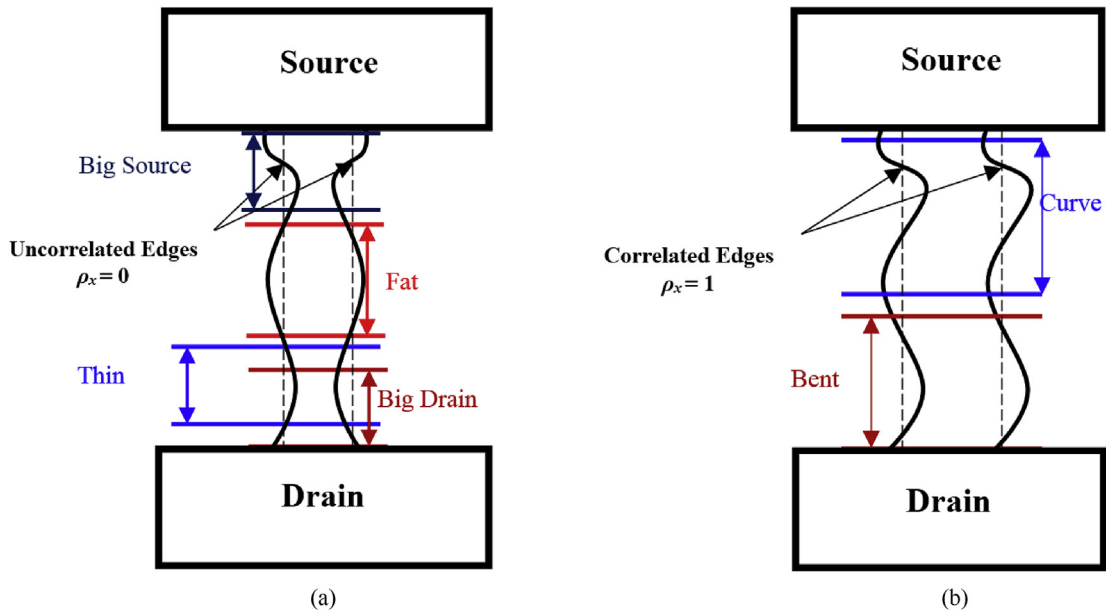


Fig. 6. Top cross-sectional view of the FinFET structure showing Fin LER with (a) resist-and (b) spacer-defined fin. Random LER on fin sides walls are correlated-and uncorrelated-in-phase for spacer-and resist-defined fins, respectively.

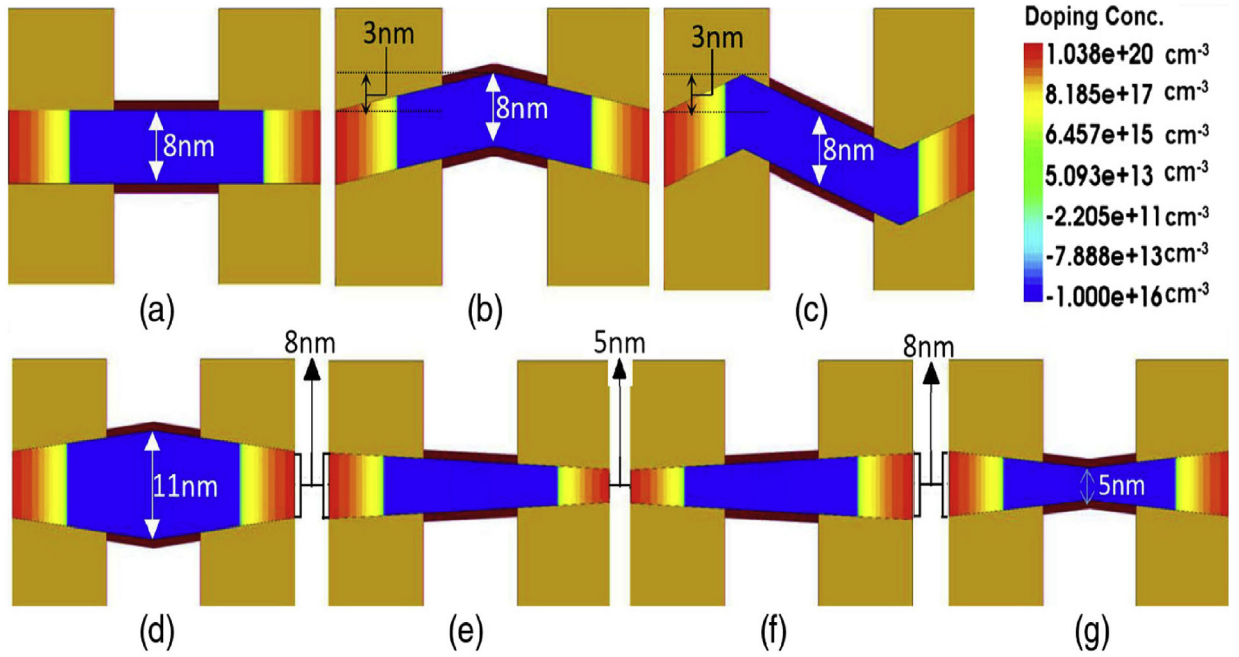


Fig. 7. Device structures and net doping profile for the distribution for all seven fin shapes (a) Rectangle (without LER) (b) bent (c) curve (d) big drain (e) big source (f) fat (g) thin.

3. Simulation result and discussion

3.1. Fin width sensitivity analysis

This sub-section describe the sensitivity of performance parameters like V_{TH} , I_{ON} , I_{OFF} and $DIBL$ as function of fin width. Here, the nominal (i.e. rectangular) FinFET structure has been considered as a reference device such that there is no offset (δ) or width differences (ΔW) among both edges of fin at source and drain end. Table 2 illustrates the DC performance parameters of nominal (i.e. rectangular) FinFET structure, which closely follows the fabricated Intel 14-nm FinFET technology node [29].

Fig. 8 plots the impact of the fin width (W_{fin}) on threshold voltage (V_{TH}) of rectangular FinFET structure to evaluate the sensitivity of V_{TH} . The simulated values of threshold voltage have been extracted by using maximum transconductance ($g_{m,max}$) method, whereas analytical values have been calculated by using statistical regression method [34]. Linear Regression method has been used to develop a relationship between dependent variable (V_{TH}) and independent variables (W_{fin}) using a best fit straight line also known as regression line. It is represented by an equation $V_{TH} = \beta_0 + \beta_1 W_{fin}$, where, “ W_{fin} ” is the predictor or regressor variable, “ V_{TH} ” is the response variable, β_0 is the intercept of the line, and β_1 is slope. This equation can be used to predict the value of target variable based on given predictor variable. β_0 and β_1 values can be predicted using Equations (16) and (17),

$$\beta_1 = \frac{\sum_{i=1}^n (W_{fin,i} - \bar{W}_{fin})(V_{TH,i} - \bar{V}_{TH})}{\sum_{i=1}^n (W_{fin,i} - \bar{W}_{fin})^2} \text{ and} \quad (16)$$

$$\beta_0 = \bar{V}_{TH} - \beta_1 \bar{W}_{fin} \quad (17)$$

where, $\bar{W}_{fin}$ is the mean (average) of the W_{fin} values and $\bar{V}_{TH}$ is the mean of V_{TH} . From Equations (16) and (17), regression fit line equation is given by,

$$V_{TH} = -10.5(W_{fin}) + 309.78 \quad (18)$$

Further, the coefficient of determination (R^2) is given by,

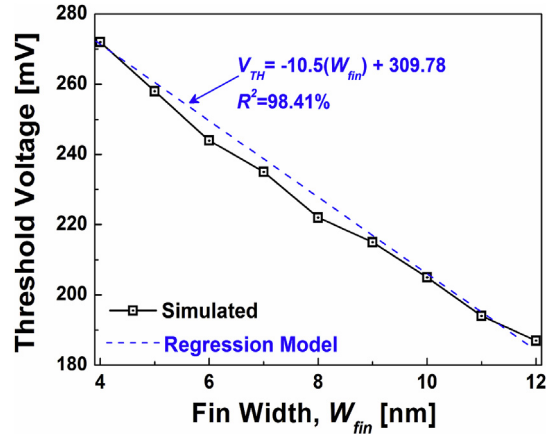


Fig. 8. Threshold voltage dependence on the fin width of FinFET structure.

$$R^2 = \left[\frac{1}{n} \sum_{i=1}^n \frac{(W_{fin,i} - \bar{W}_{fin})(V_{TH,i} - \bar{V}_{TH})}{(\sigma W_{fin})(\sigma V_{TH})} \right]^2 \quad (19)$$

here, σW_{fin} and σV_{TH} can be calculated by Equations (20) and (21),

$$\sigma W_{fin} = \sqrt{\sum_{i=1}^n (W_{fin,i} - \bar{W}_{fin})^2 / n} \quad \text{and} \quad (20)$$

$$\sigma V_{TH} = \sqrt{\sum_{i=1}^n (V_{TH,i} - \bar{V}_{TH})^2 / n} \quad (21)$$

From Equation (19), coefficient of determination is calculated as $R^2 = 0.984$ or 98.41%. It is clear that simulated and regression model values have good matches. From the curve the slope (i.e. $\partial V_{TH} / \partial W$) comes out to be -10.5 mV/nm. Therefore, the threshold voltage sensitivity can be calculated from the model Equation (15) as given below,

$$\Delta_{V_{TH}}^2 = \left(\partial V_{TH} / \partial W \right)^2 \Delta_{\delta W}^2 \quad (22)$$

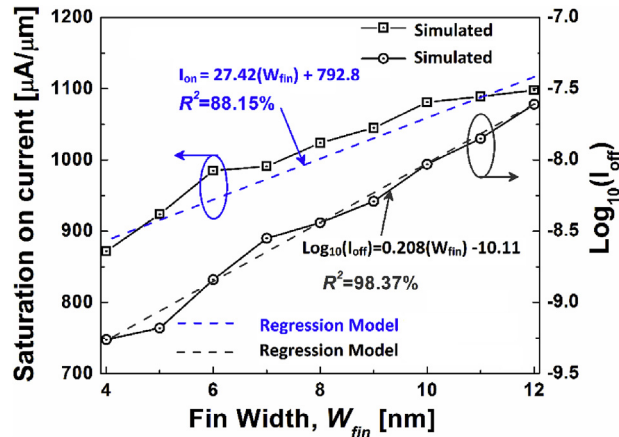


Fig. 9. On- and Off-current dependence on the fin width of FinFET structure.

Thus, by using Equation (22) the threshold voltage sensitivity is calculated as 25 mV for $\Delta_{\delta W} = 3$ nm. So, it is clear that the fin width fluctuations have significant impact on the threshold voltage sensitivity.

Similarly, Fig. 9 shows the fin width dependence over on-and off-current. Here, it is found from the figure that the slope of on (i.e. $\partial I_{ON}/\partial W$) and off [i.e. $\partial \text{Log}_{10}(I_{OFF})/\partial W$] current is 27.42 $\mu\text{A}/\text{nm}$ and 0.208 dB, respectively.

Now, model Equation (15) can be used to calculate the on-and off-current sensitivity. The sensitivity equations for on-and off-current is given as below,

$$\Delta_{I_{ON}}^2 = \left(\partial I_{ON}/\partial W \right)^2 \Delta_{\delta W}^2 \quad (23)$$

$$\Delta_{I_{OFF}}^2 = \left(\partial \text{Log}_{10} I_{OFF}/\partial W \right)^2 \Delta_{\delta W}^2 \quad (24)$$

Equations (23) and (24) calculates the sensitivity of on-and off-current as 82.25 μA and 1.87 dB for given value of for $\Delta_{\delta W} = 3$ nm. Further, similar analysis has been performed in Fig. 10 to calculate the sensitivity of *DIBL* with reference to the fin width for a given for $\Delta_{\delta W} = 3$ nm. Consequently, it is observed that the *DIBL* sensitivity comes out to be 4.67 mV/nm. Hence, this can be concluded that fin thickness variation of FinFET structure has a strong correlation (above 88% of all cases) in all performance parameters.

3.2. Impact of LER variability on FinFET structure

In this sub-section, the impact of LER has been investigated on all possible fin shapes as resulted due to the spacer- (i.e. correlated LER) and-resist (i.e. uncorrelated LER)-defined patterning techniques (Section. 2.3).

The I_D - V_{GS} characteristics of the spacer-(i.e. correlated LER) and-resist (i.e. uncorrelated LER)-defined FinFETs along with rectangular FinFET are shown in Fig. 11 (a) and 11 (b), respectively. It is observed from figure that, the spacer-defined FinFET i.e. Fig. 11 (a) closely follows the rectangular FinFET transfer characteristics, whereas there is large mismatch in case of resist-defined FinFET characteristics i.e. Fig. 11 (b). This suggests that the spacer-defined technique provides better performance.

The electrostatic potential distribution among all fin shapes has been plotted in Fig. 12 at $V_{GS} = 0.70$ V and $V_{DS} = 0.05$ V. This can be seen from Fig. 12 that, the shape of the electrostatic potential distribution under gate region is completely governed by the fin geometry, and LER variability. These fluctuations in fin geometry can change the electrostatic control of the device. This means that the magnitude of LER variability plays an important role in controlling the channel region electrostatics. So, it is clear from Fig. 12 that resist-and spacer-defined FinFETs have a different channel controllability. It has been found that, the spacer-defined FinFETs exhibits symmetrical distribution [i.e. Fig. 12(b)-(c)] of potential profile due to constant thickness of the fin as compared to the resist-defined FinFET [i.e. Fig. 12(d)-(g)] structures. These fluctuations in the potential distribution profile are responsible for variation in the threshold voltage (V_{TH}) for all the FinFET structures. Fig. 13 displays the V_{TH} value for spacer-and resist-defined FinFETs with respect to the rectangular FinFET structure. It is clear from figure that bent FinFET is showing better immunity to V_{TH} fluctuation (i.e. 1.35%) followed by a curve (i.e. 4.96%) and fat (i.e. 4.50%) FinFET structures. This is because of the fact that the variation in the potential distribution of these FinFET structure underneath the gate region is almost identical to that of rectangular FinFET structure. However, the impact of LER variability is more significant in all other fin shapes such as bigD, bigS, and thin FinFETs due to diminution of overall fin thickness under

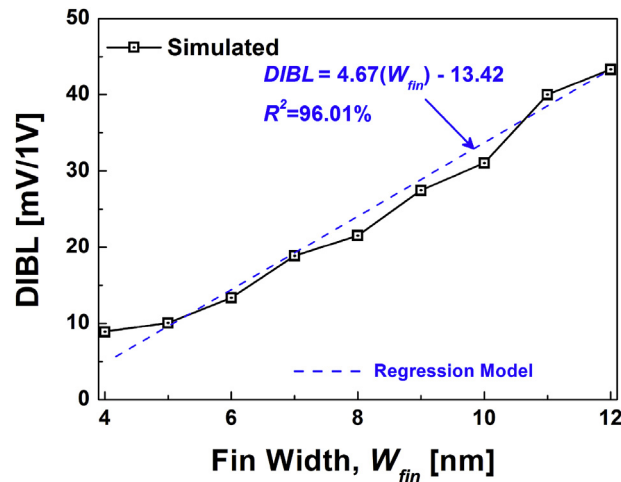


Fig. 10. DIBL dependence on the fin width of FinFET structure.

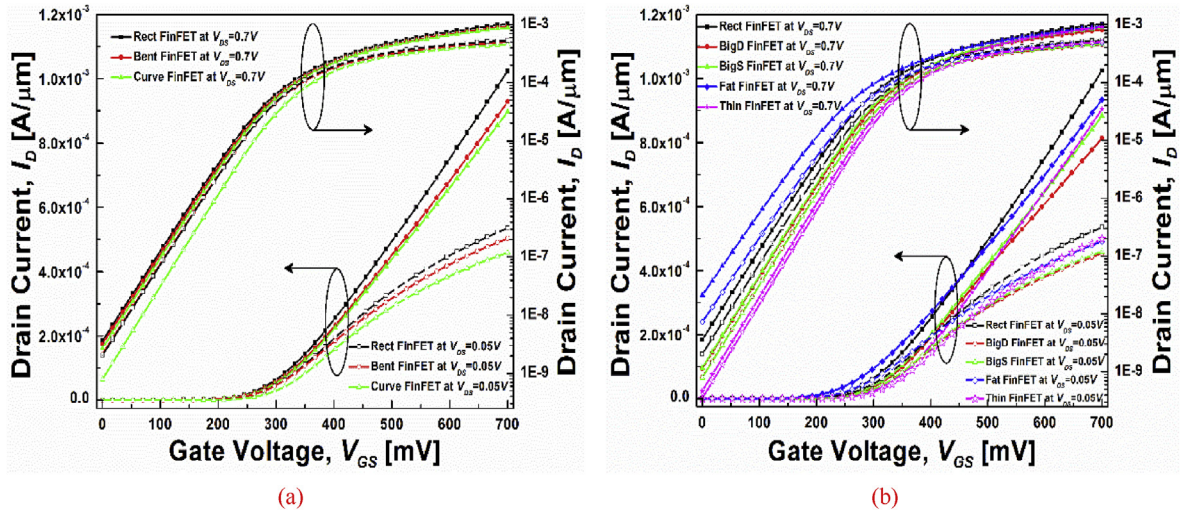


Fig. 11. I_D - V_{GS} characteristics (a) Spacer (b) Resist defined FinFETs with respect to rectangular FinFET structure.

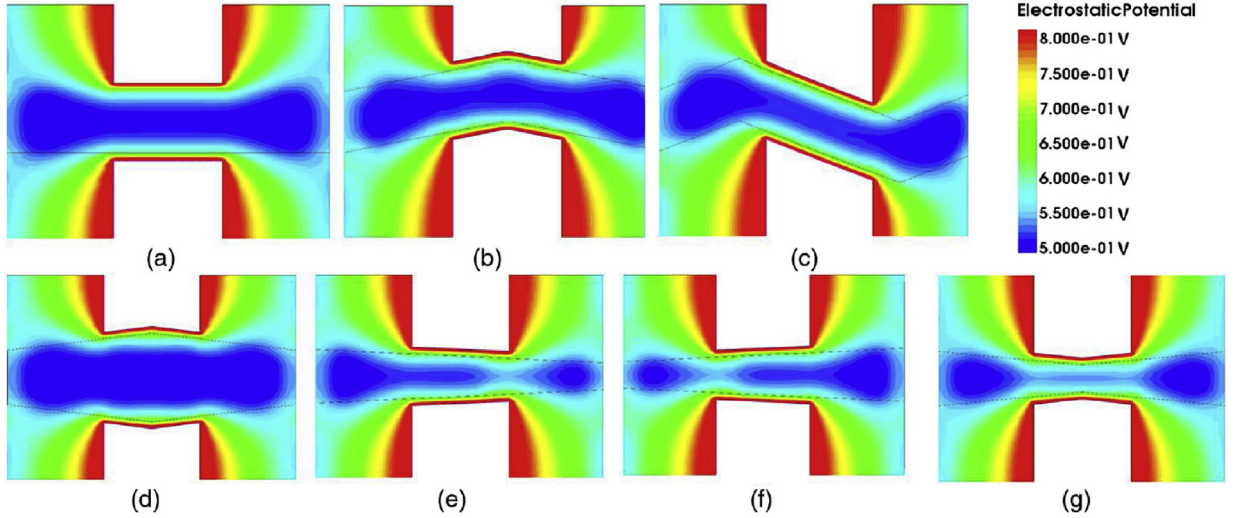


Fig. 12. Plots of on-state electrostatic potential distribution for all seven fin Shapes (a) Rectangle (without LER) (b) bent (c) curve (d) fat (e) big source (f) big drain (g) thin.

gate region. This leads to enhanced quantum confinement effects and it dominates in the thinner segment of fin which in turn increases the V_{TH} fluctuation of the device [35]. As a result, the thin FinFET demonstrates the highest V_{TH} variability 16.22% followed by bigD and bigS FinFET structures. Thus, resist-defined FinFET shows significant V_{TH} fluctuations due to the asymmetrical thickness of the fin under the gate region as compared to spacer-defined FinFETs. Fig. 14 represents the electron density distribution in different fin shapes for on-state condition. It is clearly seen from the figure that the position of highest electron density cloud depends upon the shape of fin. The electron density peaks (maximum) near the Si/SiO₂ interfaces for rectangular FinFET followed by fat, bent, and curve FinFETs, whereas for bigD, bigS, and thin FinFETs, the electron density is maximum at the center of the fin because of local fin thinning under gate region. The on-current for all the fin shapes has been plotted in Fig. 15 at $V_{GS} = V_{DS} = 0.70$ V. It is noted from the figure that the fat FinFET structure has minimum variability i.e. 8.79%, while big drain FinFET is showing maximum variability (i.e. 20.51%) with respect to rectangular FinFET. This is because the sheet resistance (R_{sh}) of extension region is inversely proportional to fin thickness (T_{Si}) [36]. Therefore, decreasing the fin width in the source end results in a higher sheet resistance in this region, which in turn reduces the effective V_{GS} and hence decreases the on-current of the device [37]. Fig. 16 shows the electron density distribution in off-state condition and Fig. 17 plots the off-current fluctuation at $V_{GS} = 0$ and $V_{DS} = 0.70$ V for all FinFET structures. It is clear from figure that the off-current leakage in fat fin shape increases up to 485.64% with respect to the rectangular FinFET structure. This is due to the fact that the

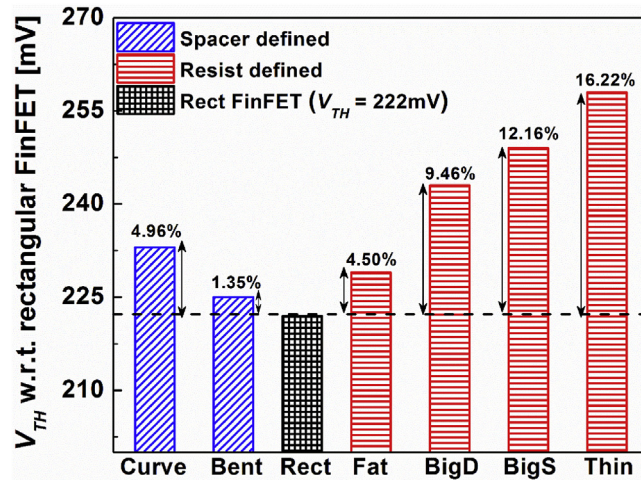


Fig. 13. Average percentage of threshold voltage fluctuations for spacer-and resist-defined FinFET structures with respect to rectangular FinFET.

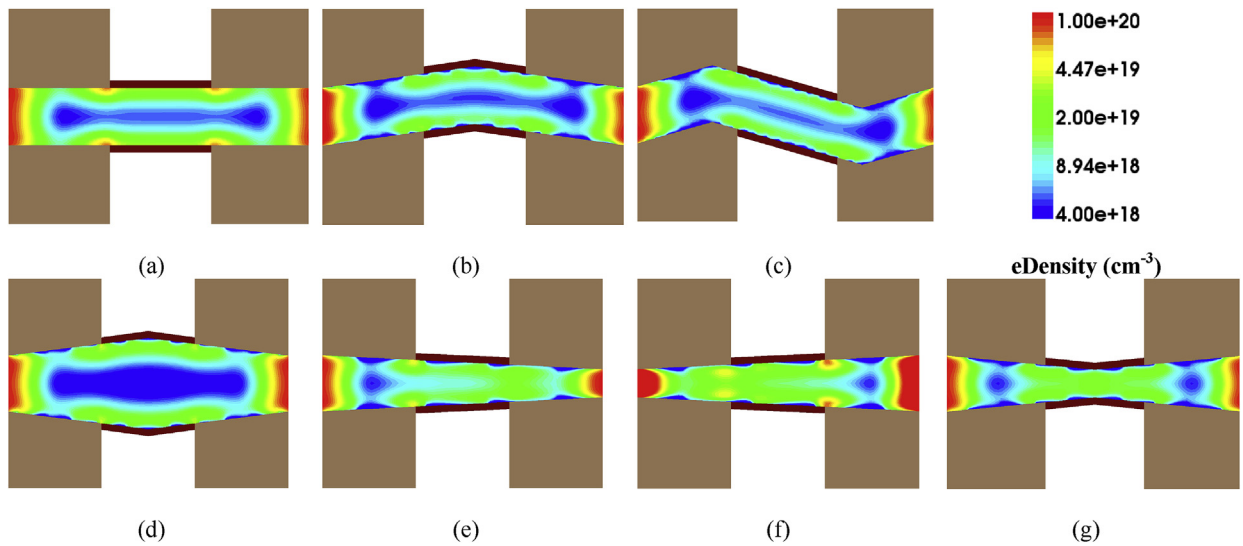


Fig. 14. Plots of on-state electron density distribution for all seven fin Shapes (a) Rectangle (without LER) (b) bent (c) curve (d) fat (e) big source (f) big drain (g) thin.

leakage current in off-state flows in the central portion of the fin irrespective of fin shape. So, the gate electrode has poor control in fat fin since the middle portion of the fin is the most remote area from the gate electrode. However, off-current leakage is 87.10% lower in thin fin with reference to the rectangular FinFET. This is due to superior control of channel region by the gate electrode. Fig. 18 shows the impact of LER variability on Drain Induced Barrier Lowering (DIBL) for spacer-and resist-defined FinFETs structures with respect to rectangular FinFET structure. This can be observed from the figure that thin FinFET improves the DIBL from 20.98 mV/V down to a reputable 13.33 mV/V. This is due to fact that, for thinner fin width devices the gate control in channel improves significantly. Moreover, the fat fin shape exhibits the worst DIBL performance by increasing its affect up to 119.11% with respect to rectangular FinFET structure.

Fig. 19 summarizes the average percentage fluctuations in electrical parameters for spacer-and resist-defined FinFET structures with respect to rectangular FinFET structure. It is concluded that LER variability impact is significantly lower in spacer-defined FinFET structures. This is because of the fact that fin thickness between both edges fluctuates in a correlated manner such that average fin thickness remains nearly the same as that of rectangular FinFET. As a result, spacer-defined FinFET structures have reduced parameter fluctuations.

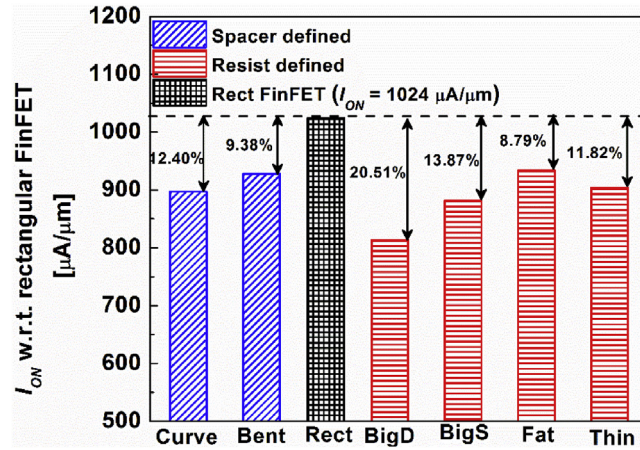


Fig. 15. Average percentage of on-current fluctuations for spacer-and resist-defined FinFET structures with respect to rectangular FinFET.

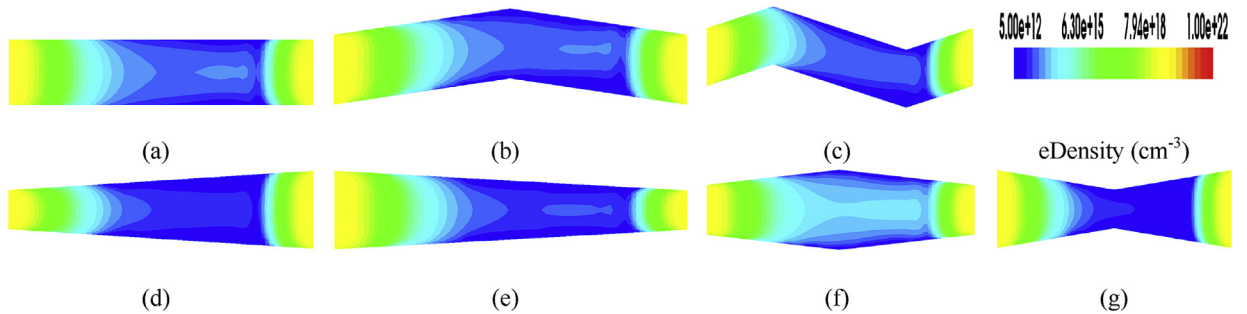


Fig. 16. Plots of off-state electron density distribution for all seven fin Shapes (a) Rectangle (without LER) (b) bent (c) curve (d) big drain (e) big source (f) fat (g) thin.

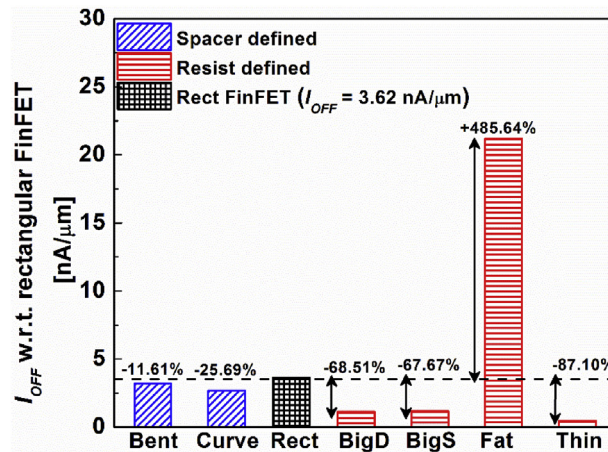


Fig. 17. Average percentage of off-current fluctuations for spacer-and resist-defined FinFET structures with respect to rectangular FinFET.

3.3. LER variability impact on SRAM – a circuit implication

Variations of FinFETs electrical parameters due to LER variability affect the SRAM operations as it is quite sensitive to threshold voltage and drive current fluctuations between transistor pairs [22–24]. A circuit schematic of conventional six transistor (6-T) SRAM cell is shown in Fig. 20, which consists of two cross coupled inverters, two-bit lines, and a single word line. From Fig. 20, T1 and T3 are *n*-channel FinFET pull-down (PD) transistors, T2 and T4 are *p*-channel FinFET pull-up (PU)

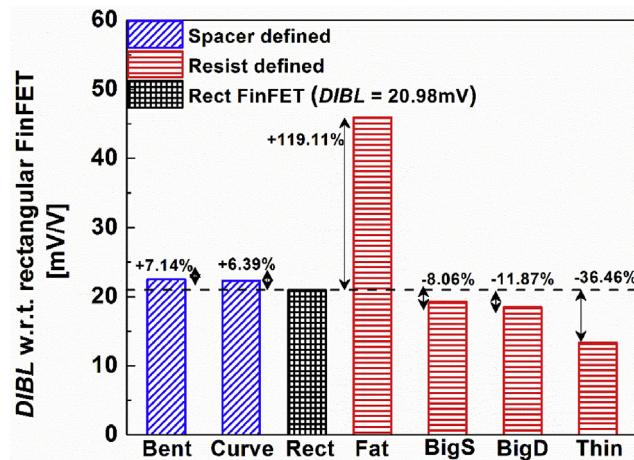


Fig. 18. Average percentage of DIBL fluctuations for spacer-and resist-defined FinFET structures with respect to rectangular FinFET.

transistors, and T5 and T6 are n-channel FinFET access transistors (AC). The output data bits generated are stored in the nodes Q and Q'. The signal-to-noise margin (SNM) is one of the most crucial parameter that characterizes the stability of SRAM cell. SNM can be defined as the maximum value of dc noise voltage that is accepted without altering stored bit, and it is calculated by the fitting a square between the two voltage transfer characteristics (VTC) curves of cross coupled cell inverters. In the present work, a full 3-D TCAD mixed mode simulation has been performed to investigate the impact of fin-LER variability on the 6-T SRAM cell performance such as fluctuation in read static noise margin (RSNM). FinFET based SRAMs cell resulted from the spacer-and resist-defined techniques have also been explored to assess the stability of FinFETs by knowing the static noise margin (RSNM) in each case.

RSNM measures SRAM's stability against the unwanted state flipping during read operation. The butterfly curve method has been considered in order to evaluate the SNM. Fig. 21 demonstrates the variation in butterfly voltage transfer characteristics (VTC) curves for read SNM for spacer-and resist-defined FinFETs with respect to rectangular FinFET structure. It is noted from the butterfly curves that average RSNM for spacer-and resist-defined FinFETs are 355 mV and 295 mV, respectively at $V_{DD} = 0.70$ V. Whereas for rectangular FinFET structure RSNM is found to be 385 mV. Thus, it is clear from the figure that the RSNM of conventional 6-T SRAM bit cell degrades drastically under the influence of LER variability. This is due to the fact that LER variability directly affects the SNM by modulating the effective fin thickness. The effective RSNM of spacer-and resist-defined individual FinFETs have been shown in Fig. 22. It is observed from Fig. 22 that percentage degradation in read static noise margin (RSNM) due to LER variability in curve, bent, big drain, big source, fat and thin fin shapes with respect to rectangular fin shape is 5.19%, 7.01%, 19.48%, 20.77%, 13.50% and 24.68%, respectively. The RSNM variations in spacer-defined fin shapes are lesser as compared to resist-defined fin shapes. This is mainly because in resist-defined FinFETs the average fin thickness is not uniform which leads to large V_{TH} and I_{ON} variations, thereby degrading the performance of the driver transistors in the SRAM cell. Whereas in spacer-defined FinFETs the average thickness remains constant, so there is less

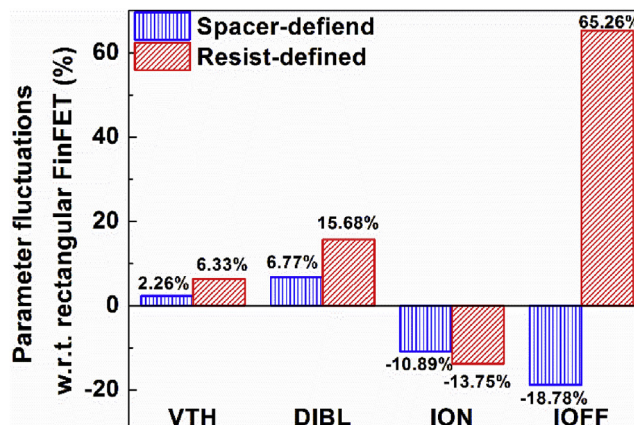


Fig. 19. Average percentage of all electrical parameters fluctuations for spacer-and resist defined FinFET structures with respect to rectangular FinFET.

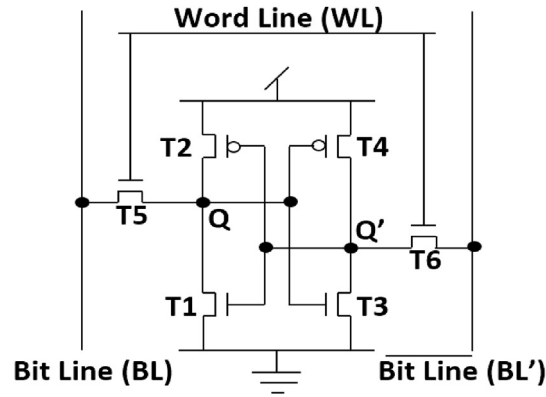


Fig. 20. Circuit schematic of a six transistor SRAM cell.

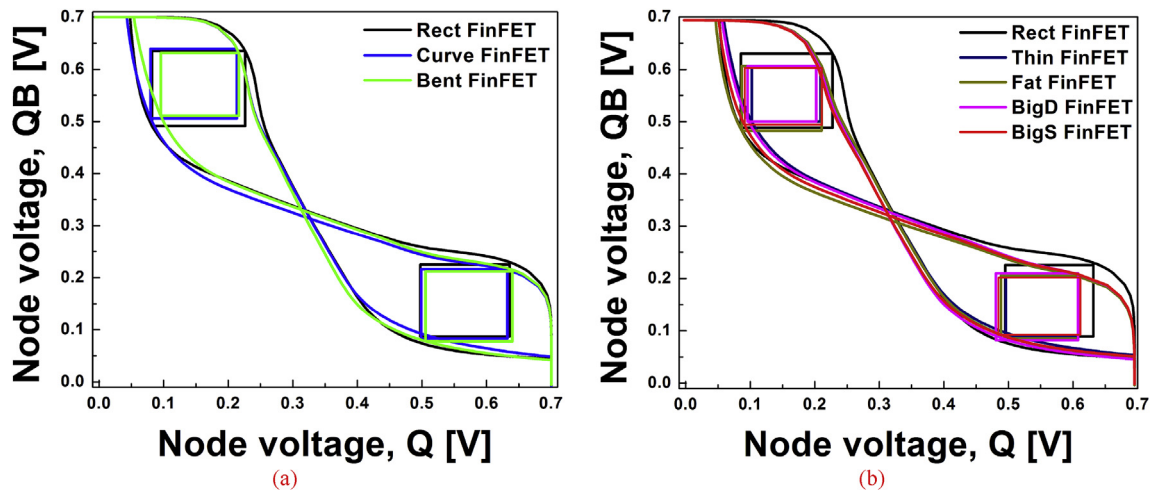


Fig. 21. Butterfly curves for read static noise margin (RSNM) (a) Spacer (b) Resist defined FinFETs.

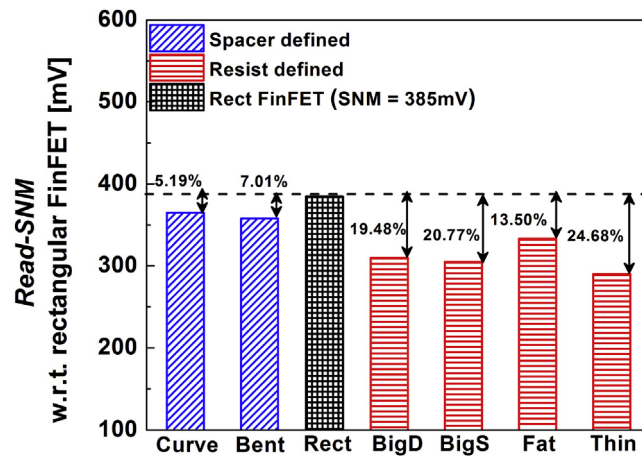


Fig. 22. RSNM for spacer-and resist-defined FinFET structures with respect to rectangular FinFET.

variation in V_{TH} and I_{ON} . Therefore, at sub-20 nm technology nodes, LER induced random fluctuations are one of the major technical barriers that must be overcome while developing the next-generation FinFET technology and to circumvent this problem to some extent spacer-defined patterning is the best possible solution.

4. Conclusion

In this paper, a 2-D analytical model of fin-1ER is proposed to differentiate between the fin shape fluctuations for resist- and spacer-defined FinFETs. The impact of 1ER variability has been studied in all possible shapes of fin to know the effect of 1ER on the performance of spacer- and resist-defined FinFETs structures. It is found that fin shape fluctuations in FinFET structure lead to current crowding and deteriorates the performance of the device. Consequently, we can conclude that, as compared to resist-defined FinFETs, the spacer-defined FinFETs shows least fluctuation and closest nominal performance to the rectangular FinFET structure. Therefore, spacer-defined patterning is the most promising fabrication technology for transistor scaling to the end of the technology roadmap.

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